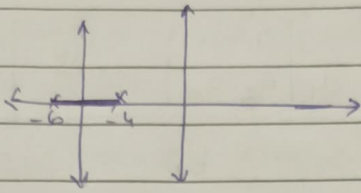


Assignment - 4

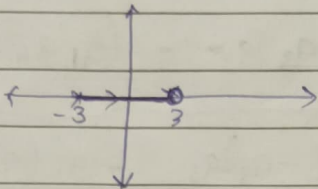
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Q1. a)



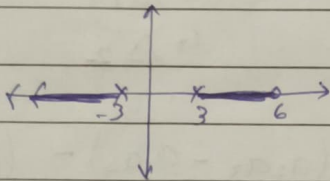
$$G(s) = \frac{1}{(s+4)(s+6)}$$

b)



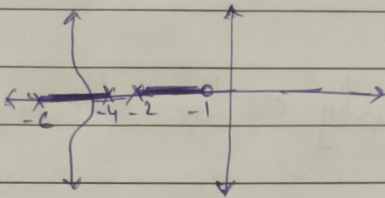
$$G(s) = \frac{s-3}{s+3}$$

c)

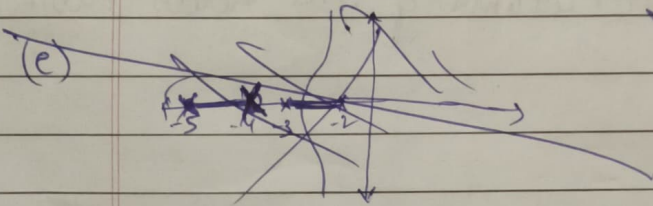


$$G(s) = \frac{s-6}{(s-3)(s+3)}$$

d)

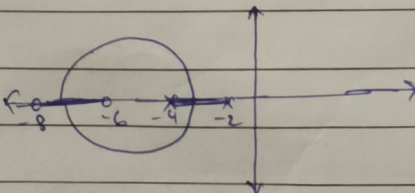


$$G(s) = \frac{s+1}{(s+2)(s+4)(s+6)}$$



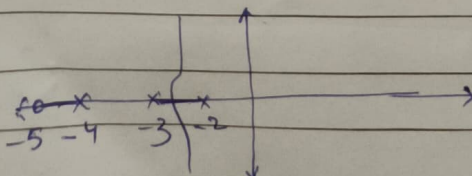
$$G(s) = \frac{(s+4)}{(s+2)(s+3)(s+5)}$$

f)



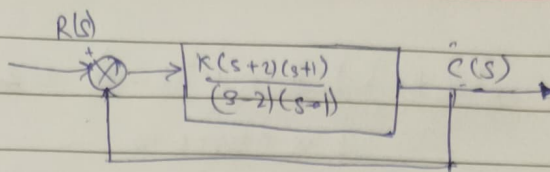
$$G(s) = \frac{(s+6)(s+8)}{(s+4)(s+2)}$$

e)

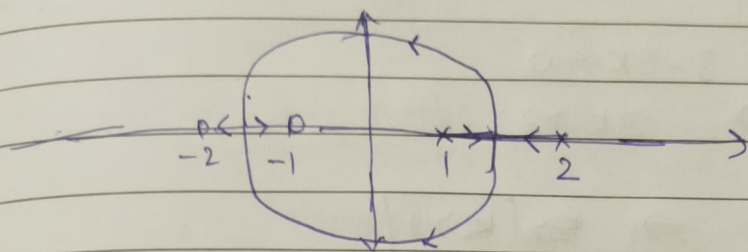


$$G(s) = \frac{(s+5)}{(s+2)(s+3)(s+4)}$$

Q2. System 1:-



$$G(s) = \frac{(s+2)(s+1)}{(s-2)(s-1)}$$



a) for break-away break-in points:

$$K = -1/G(s) = - \frac{(s-2)(s-1)}{(s+2)(s+1)} \Rightarrow \frac{(s^2 - 3s + 2)}{(s^2 + 3s + 2)}$$

$$\frac{dK}{ds} = \frac{(2s-3)(s^2+3s+2) - (2s+3)(s^2-3s+2)}{(s^2+3s+2)^2}$$

$$\cancel{2s^3} + \cancel{6s^2} + \cancel{4s} - \cancel{3s^2} - \cancel{9s} - 6 - [\cancel{2s^3} - \cancel{6s^2} + \cancel{4s} + \cancel{3s^2} - \cancel{9s} + 6]$$

$$6s^2 - 12 = 0$$

$$s = \pm \sqrt{2}$$

Clearly

break-away point = $\sqrt{2}$

break-in point = $-\sqrt{2}$

b) For system to be stable poles must be in the LHP.

i.e. $(s-2)(s-1) + K(s+1)(s+2) = 0$ } $\rightarrow$ roots in LHP.

$$\Rightarrow s^2(1+K) + s(3K-3) + 2(K+1) = 0$$

$$\text{if } as^2 + bs + c = 0$$

Now for roots to lie in LHP, $-b/a < 0$, $c/a > 0$

$$\Rightarrow \frac{2(K+1)}{(K+1)} > 0 \quad \text{and} \quad \frac{3-3K}{1+K} < 0, \quad \text{Now as } K > 0 \Rightarrow 1+K > 0$$

$$\Rightarrow \therefore 3-3K < 0$$

$$\Rightarrow 3K > 3$$

$$\Rightarrow \boxed{K > 1}$$

$\therefore K > 1$ for stable system.

(c) for critically damped poles

$$s^2(K+1) + 3s(K+1) + 2(K+1) = (s-x)^2$$

$$\text{or } s^2 + 3s\left(\frac{K+1}{K+1}\right) + 2 = (s-x)^2 \quad \text{--- (1)}$$

$$\Rightarrow 3\left(\frac{K+1}{K+1}\right) = 2\sqrt{2}$$

$$3K - 3 = 2\sqrt{2}K + 2\sqrt{2}$$

$$\Rightarrow K(3 - 2\sqrt{2}) = 2\sqrt{2} + 3$$

$$\Rightarrow \boxed{K = \frac{2\sqrt{2} + 3}{3 - 2\sqrt{2}}} = 33.97$$

as $K > 1$ the system is stable.

d) ~~Star~~ Compare the eqn with std. 2nd order eq.
i.e. $s^2 + 2\zeta\omega_n s + \omega_n^2$

$$\Rightarrow 2\zeta\omega_n = \frac{3(K-1)}{(K+1)}, \quad \omega_n^2 = 2$$

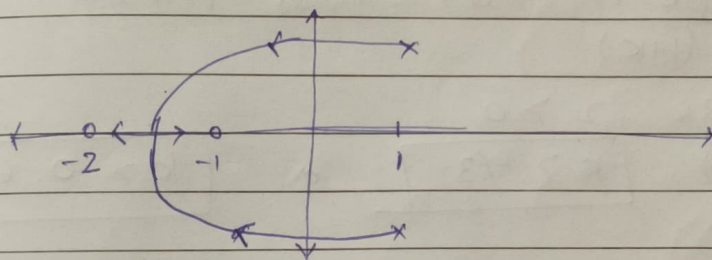
Now $\zeta = 1/\sqrt{2} = 0.707$, $\omega_n = \sqrt{2}$

$$\Rightarrow 2 = 3 \left(\frac{K-1}{K+1} \right)$$

$$2K + 2 = 3K - 3$$

$$\boxed{K = 5}$$

System 2: $K G(s) = \frac{K(s+1)(s+2)}{(s^2 - 2s + 2)}$
 $\rightarrow$ roots at $1+j$, $1-j$



a) No break-away points, for break-in points

$$K = -1/G(s) = -\frac{(s^2 - 2s + 2)}{s^2 + 3s + 2}$$

$$\frac{dK}{ds} = - \frac{[(2s-2)(s^2+3s+2) - (2s+3)(s^2-2s+2)]}{(s^2+3s+2)^2}$$

$$2s^3 + 6s^2 + 4s - 2s^2 - 6s - 4 - [2s^3 - 4s^2 + 4s - 6s + 3s^2 + 6]$$

$$5s^2 - 10 = 0$$

$$s = \pm \sqrt{2}$$

$$\Rightarrow \boxed{\text{break-in point at } -\sqrt{2}}$$

b) For stable system roots of

$$s^2 - 2s + 2 + k(s+2)(s+1) \text{ lie in LHP}$$

$$\hookrightarrow s^2(1+k) + s(3k-2) + 2(k+1) = 0$$

For roots in LHP $\frac{2(k+1)}{(k+1)} > 0 \Rightarrow 2 > 0 \checkmark$

$$- \frac{(3k-2)}{(1+k)} < 0$$

$$\Rightarrow \frac{(3k-2)}{(1+k)} > 0, \text{ as } 1+k > 0 \therefore k > 0$$

$$3k - 2 > 0$$

$$\Rightarrow \boxed{k > 2/3} \quad \alpha \quad \boxed{k > 0.667}$$

c) for critically damped poles

$$s^2 + s \left(\frac{3k-2}{k+1} \right) + 2 = (s-\alpha)^2 \quad - (2)$$

$$\Rightarrow \frac{3k-2}{k+1} = 2\sqrt{2}$$

$$\Rightarrow 3k - 2 = 2\sqrt{2}k + 2\sqrt{2}$$

$$K = \frac{2(1+\sqrt{2})}{3-2\sqrt{2}} \quad \text{or} \quad K \geq 28.142$$

as $K > 2/3$ the system is stable.

d) compare eq (2) with std. 2nd order eq.

$$s^2 + s \left(\frac{3K-2}{K+1} \right) + 2 = s^2 + 2\zeta\omega_n s + \omega_n^2$$

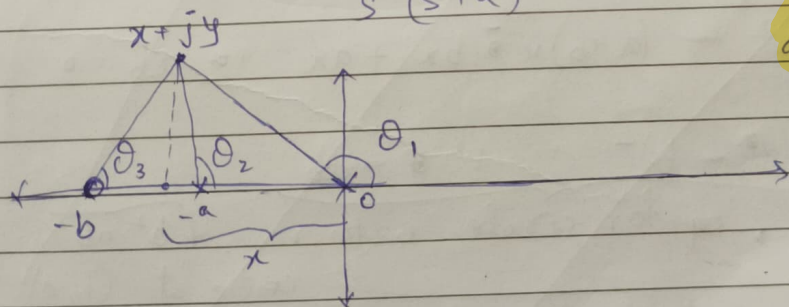
for $\zeta = 1/\sqrt{2}$ or 0.707, here $\omega_n^2 = 2$

$$\Rightarrow 2 \times \frac{1}{\sqrt{2}} \times \sqrt{2} = \frac{3K-2}{K+1}$$

$$\Rightarrow K = 4$$

Q3 $G(s)(H(s)) = \frac{K(s+b)}{s(s+a)}$

assuming $b > a$
as root locus is
a st. line for $a > b$.



Take a point $x+jy$ on the root locus, now

$$\theta_3 - \theta_1 - \theta_2 = (2n+1)\pi$$

$$\Rightarrow \tan^{-1} \left(\frac{y}{b+x} \right) - [\pi - \tan^{-1} (y/x)] - [\pi - \tan^{-1} (y/x-a)] = (2n+1)\pi$$

as x is negative.

$$\Rightarrow \tan^{-1}\left(\frac{y}{-x}\right) + \tan^{-1}\left(\frac{y}{-x-a}\right) + \tan^{-1}\left(\frac{y}{b+x}\right) = (2n+1)\pi + 2\pi$$

$$* \quad \tan^{-1}a + \tan^{-1}b + \tan^{-1}c = \tan^{-1}\left(\frac{a+b+c - abc}{1 - ab - bc - ca}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{y}{-x} + \frac{y}{-x-a} + \frac{y}{b+x} - \frac{y^3}{-x(-x-a)(b+x)}}{1 - \frac{y^2}{x(-x-a)} - \frac{y^2}{x(b-x)(x-a)} - \frac{y^2}{x(b+x)}}$$

Take tan on both sides

$$\Rightarrow \frac{y(-x-a)(b+x) + y(-x)(b+x) + y(-x)(-x-a) - y^3}{-x(-x-a)(b+x) - y^2(b+x) - y^2(-x) - y^2(-x-a)} = 0$$

$$\Rightarrow -x^2 + (a+b)x - ab + xb - x^2 + x^2 - xa - y^2 = 0$$

$$y^2 + x^2 - (a+b)x - bx + ax + ab = 0$$

$$y^2 + (x^2 - b^2) = b^2 - ab$$

eq for circle with $r = \sqrt{b^2 - ab}$, and centre at $(b, 0)$

$$\Rightarrow y^2 + x^2 + (a+b)x + ab + bx + x^2 - x^2 - xa = 0$$

$$y^2 + x^2 + 2bx + ab = 0$$

$$\Rightarrow y^2 + (x+b)^2 = b^2 - ab$$

eq of circle with $r = \sqrt{b^2 - ab}$ & centre at $(-b, 0)$

$$a = 4, \quad b = 6, \quad \Rightarrow \quad \frac{K G(s)}{1 + G(s)} = \frac{K(s+6)}{s(s+4) + K(s+6)}$$

$$s^2 + (4+K)s + 6K = 0$$

for underdamped systems ~~poles~~ poles at $-\sigma \pm j\omega_d$

$$\Rightarrow (4+K)^2 - 6K \times 4 < 0$$

$$\Rightarrow K^2 + 8K + 16 - 24K < 0$$

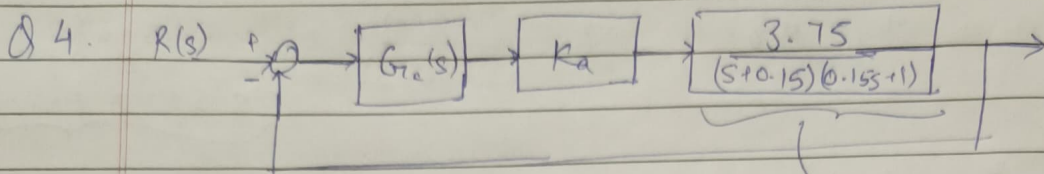
$$K^2 - 16K + 16 < 0$$

$$\Rightarrow (K-8)^2 - 48 < 0$$

$$\Rightarrow (K - (8+4\sqrt{3}))(K - (8-4\sqrt{3})) < 0$$

$\therefore$ Range of K is

$$8-4\sqrt{3} < K < 8+4\sqrt{3}$$



$$\% OS = 10\%$$

$$\Rightarrow e^{-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}} \times 100 = 10$$

$$e^{\frac{\pi \zeta}{\sqrt{1-\zeta^2}}} = 10$$

$$\frac{\pi^2 \zeta^2}{1-\zeta^2} = 5.302$$

$$\frac{\zeta^2}{1-\zeta^2} = 0.5372$$

$$\zeta^2 = 0.349$$

$$\boxed{\zeta = 0.591}$$

$$G(s) = \frac{3.75}{(s+0.15)(0.15s+1)}$$

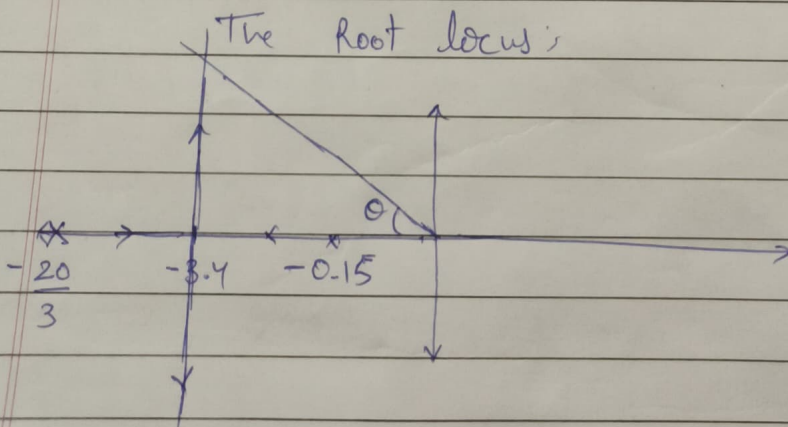
$$= \frac{25}{(s+0.15)(s+20/3)}$$

$$T < 1.5$$

$$\frac{4}{\zeta \omega_n} < 1.5$$

$$\zeta \omega_n > \frac{4}{1.5} = 2.6667$$

The Root locus:



$$\theta = \cos^{-1} \zeta$$

Now as $\zeta \omega_n = 3.4$, the settling time criteria is already satisfied.

Now $\zeta\omega_n = 3.9$, $\rho = 0.591$, $\Rightarrow \omega_n^m = 5.767$

Now the transfer function. $\frac{K G_c(s)}{1 + K G_c(s)} = \frac{N(s)}{D(s)}$

$$D(s) = (s + 0.15)(s + 20/3) + 25K$$

$$\Rightarrow s^2 + 6.81s + 1 + 25K = s^2 + 2\zeta\omega_n s + \omega_n^2$$

$$\Rightarrow 1 + 25K = (5.767)^2, \Rightarrow \boxed{K = 1.29}$$

Now for the proportional plus integral compensator, we need to add a pole at origin to make steady state error zero for a step command. Also to compensate for the pole at ~~the~~ origin, we ~~defi~~ also add a zero very close to the origin so that ~~no~~ our root locus remains approx. same.

$\therefore$ Final design:- $G_c(s) = \frac{s + 0.00001}{s}$

$$\boxed{K_a = 1.29}$$

Q5 $G(s) = \frac{K}{s(s+10)(s+50)}$

$\%OS = 7.5\%$

$e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100 = 7.5$

$\frac{\pi\zeta}{\sqrt{1-\zeta^2}} = 13.33$

$\frac{\zeta}{\sqrt{1-\zeta^2}} = 0.824$

$\zeta^2 = 0.404$

$\zeta = 0.636$

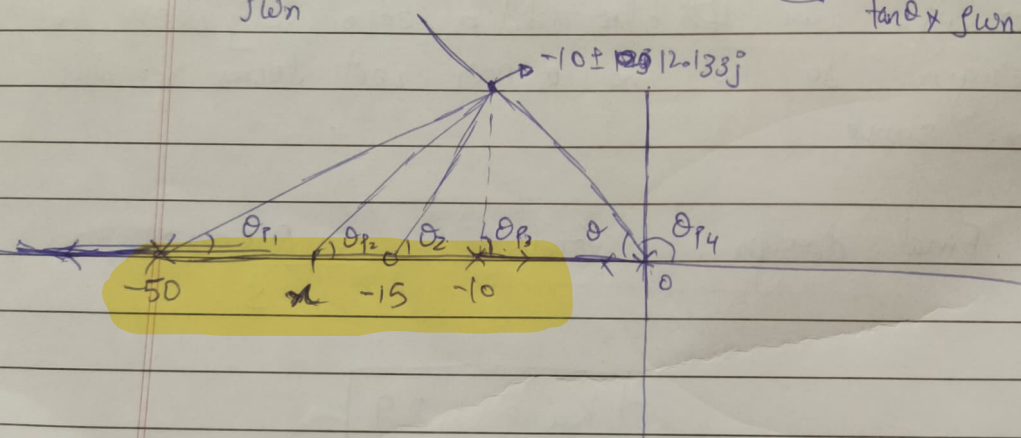
$\theta = 50.505^\circ$

$\frac{4}{\zeta\omega_n} = 0.400$

$\omega_n \geq 10$

poles at $-10 \pm 12.13j$

$\tan\theta \times \omega_n = 12.133$



Take compensator pole at $-x$

$\sum \theta_z - \sum \theta_p = (2n+1)\pi$

$\tan^{-1}\left(\frac{12.13}{5}\right) - \left[\frac{(\pi + \pi)}{2} - \tan^{-1}\left(\frac{12.133}{10}\right) + \tan^{-1}\left(\frac{12.133}{40}\right) + \tan^{-1}\left(\frac{12.133}{x-10}\right) \right] = (2n+1)\pi$

$$-168.77^\circ - \tan^{-1}\left(\frac{12.13}{x-10}\right) = \underbrace{(2n+1)\pi}_{\approx -180}$$

$$\Rightarrow \tan^{-1}\left(\frac{12.13}{x-10}\right) = 11.23^\circ$$

$$\Rightarrow \frac{12.13}{x-10} = 0.1985$$

$$x-10 = 61.09$$

$$\boxed{x = 71.09} \quad \therefore \text{ ~~compensator~~ compensator pole at } -71.09$$

$$\text{Now } K G(s) \cdot G_c(s) = \frac{K(s+15)}{s(s+10)(s+50)(s+71.09)}$$

$$K = \frac{-1}{G(s)} \quad \text{at } s = -10 \pm 12.13j$$

$$\Rightarrow K = \frac{-1}{G(-10 \pm 12.13j)} = 37837.54$$

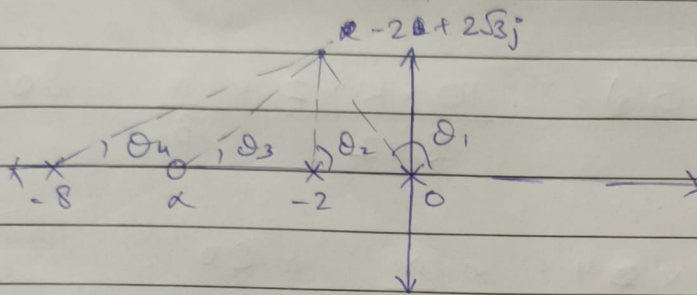
$$K_v = \lim_{s \rightarrow 0} s G(s) = \frac{37837.54(15)}{10 \times 50 \times 71.09} = 15.9674$$

$$\boxed{K_v = 15.967}$$

Q6 $G(s) = \frac{10}{s(s+2)(s+8)}$, First we apply a PD compensator.

for the poles to be at $s = -2 \pm j2\sqrt{3}$

$$G_D = K(s+x)$$



$$\theta_3 - (\theta_1 + \theta_2 + \theta_4) = (2n+1)180^\circ$$

$$\tan^{-1}\left(\frac{2\sqrt{3}}{x-2}\right) = \left[\overbrace{180 - \tan^{-1}\left(\frac{2\sqrt{3}}{2}\right)}^{60} + 90 + \overbrace{\tan^{-1}\left[\frac{2\sqrt{3}}{0}\right]}^{30} \right]$$

$$= (2n+1)180^\circ$$

$$\tan^{-1}\left(\frac{2\sqrt{3}}{x-2}\right) - 240 = -180^\circ$$

$$\frac{2\sqrt{3}}{x-2} = \tan 60 = \sqrt{3}$$

$$\boxed{x=4}$$

$$K = \frac{-1}{G(s)} = \frac{s(s+2)(s+8)}{10(s+4)} \bigg|_{s=-2+j2\sqrt{3}}$$

$K = 2.4$, Now we apply a lag compensator

with $G_c(s) = \frac{s+\alpha}{s+\beta}$, with poles and zeroes very

close to origin but in the ratio such that the kv requirements are met.

$$K_v = \lim_{s \rightarrow 0} s G(s) = \frac{10 \times K \times 2 \times \alpha}{8 \times 2 \beta} = 80$$

$$\frac{10 \times 2.4 \times 4}{8 \times 2} \times \frac{\alpha}{\beta} = 80$$

$$\Rightarrow \frac{\alpha}{\beta} = 13.33$$

$$\text{Take } \alpha = 0.0000133$$

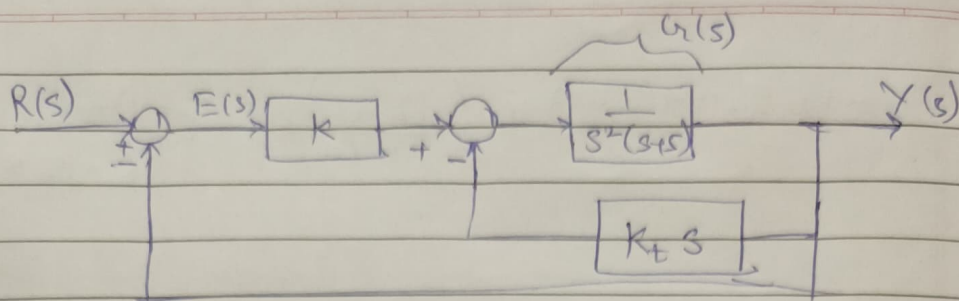
$$\beta = 0.000001$$

$\therefore$ The Final design for compensator is :-

$$G_c(s) = K \frac{(s+2)(s+\alpha)}{(s+\beta)}$$

$$= \frac{2.4 (s+4)(s+0.0000133)}{(s+0.000001)}$$

Q7.

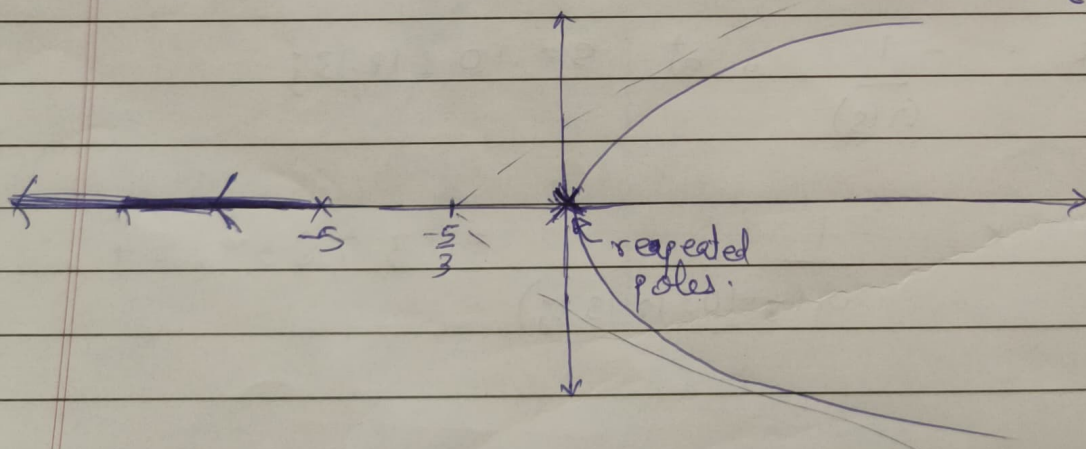


$$[(R(s) - Y(s))K - Y(s)K_t s] G(s) = Y(s)$$

$$R(s) K G(s) - Y(s) [K + K_t s] G(s) = Y(s)$$

$$\Rightarrow \boxed{\frac{Y(s)}{R(s)} = \frac{K G(s)}{1 + (K + K_t s) G(s)}}$$

a) set $K_t = 0$, characteristic eq. $1 + K \underbrace{G(s)}_{\frac{1}{s^2(s+5)}} = 0$



Total ~~asy~~ asymptotes = 3, meeting point = $-\frac{5}{3}$, angle = $\pi/3, \pi, 5\pi/3$.

~~break away point~~

b) set $K=10$, characteristic eq. $1 + (10+K)s^2 G(s) = 0$

$$T(s) = \frac{10}{s^2(s+5) + 10 + Ks}$$

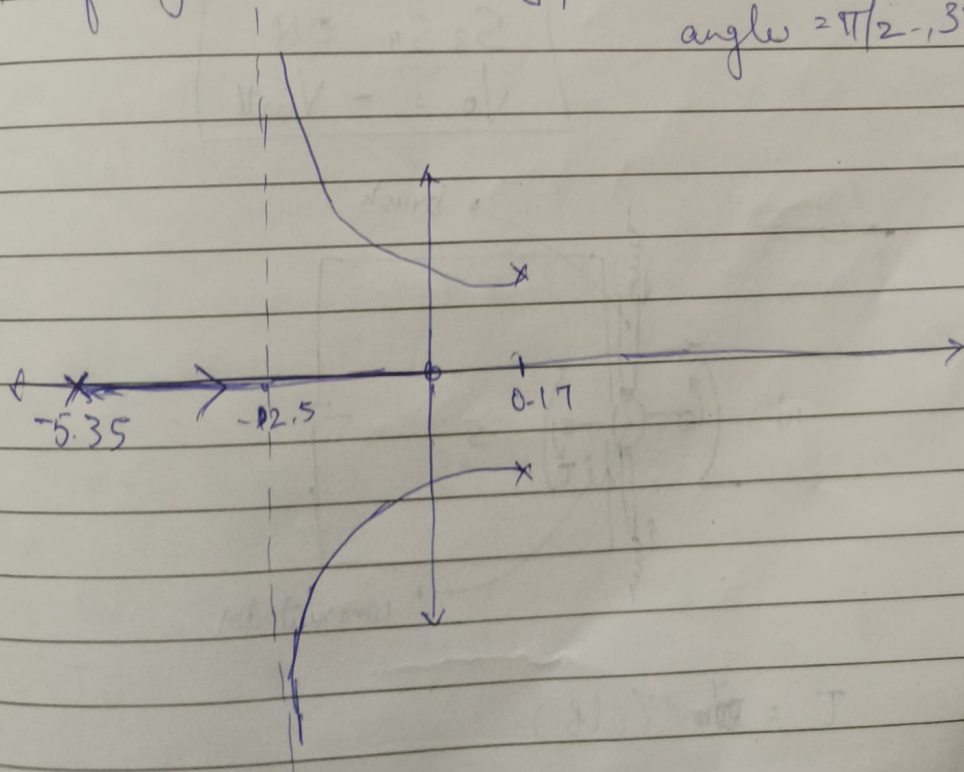
divide by $s^3 + 5s^2 + 10$

$$T(s) = \frac{10/s^2(s+5) + 10}{1 + \frac{Ks}{s^3 + 5s^2 + 10}}$$

characteristic eq.

$$G(s) = \frac{s}{s^3 + 5s^2 + 10} = \frac{s}{(s + 5.35)(s - (0.17 + 1.36j))(s - (0.17 - 1.36j))}$$

No. of asymptotes = 2, Meeting point = -2.505
angle = $\pi/2, 3\pi/2$



$$088 \quad G(s) = \frac{(K_p + K_I/s) (48500)}{(s^2 + 2.89s)}$$

$$a) \quad e_{para} = \lim_{s \rightarrow 0} \frac{1}{s^2 G(s)} = 2$$

$$= \lim_{s \rightarrow 0} \frac{(s^2 + 2.89s) \times \cancel{s}}{48500 (K_p s + K_I) \cancel{s}} = 2$$

$$= \frac{2.89}{48500 \times K_I} = 2$$

$$\boxed{K_I = 2.979 \times 10^{-5}}$$

$$b) \quad T(s) = \frac{(K_p + K_I/s) \times 48500}{s^2 + 2.89s + \left(\frac{K_I}{s} + K_p\right) 48500}$$

$$= \frac{\cancel{(K_p s + K_I)} \cancel{48500}}{\cancel{s^3 + 2.89s^2 + (K_I + K_p s) 48500}}$$

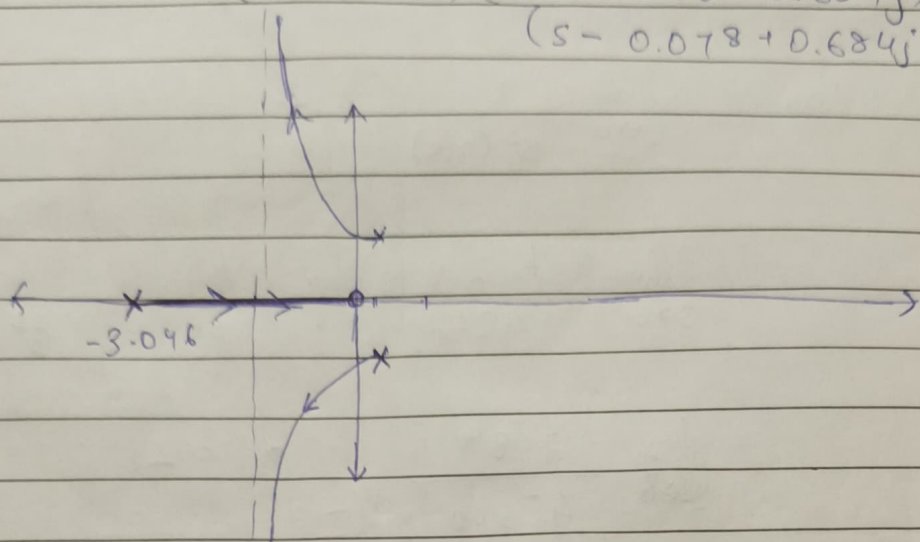
$$= \frac{N(s)}{s^3 + 2.89s^2 + (K_I + K_p s) 48500}$$

$$= \frac{N(s)}{s^3 + 2.89s^2 + \underbrace{K_p s + 1.445}_{\leftarrow \times 48500}}$$

$$= \frac{N(s)}{s^3 + 2.89s^2 + 1.445}$$

$$1 + \frac{K_p \left(\frac{s \times 48500}{s^3 + 2.89s^2 + 1.445} \right)}{L(s)}$$

$$G(s) = \frac{s \times 48500}{(s+3.046)(s-0.078-0.684j)(s-0.078+0.684j)}$$



$$\text{Meeting point of asymptotes} = \frac{-3.046 + 2 \times 0.078}{2} = 1.445$$

~~$$(c) K_p = \frac{-1}{G(0)} = \frac{-1(s+3.046)(s-0.078-0.684j)(s-0.078+0.684j)}{s \times 48500}$$~~

~~$$K_p = -1.099 \times 10^{-4}$$~~

Assuming real root at -1.

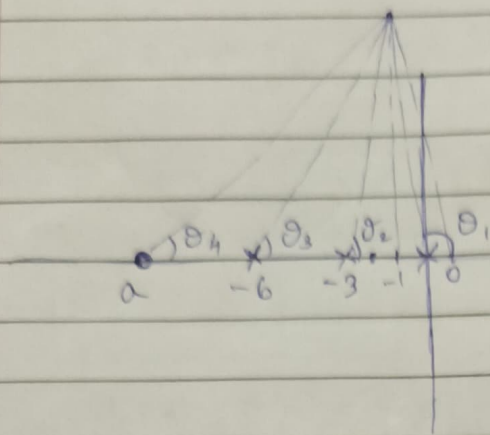
$$K_p = \frac{-1}{G(-1)} = 6.876 \times 10^{-5}$$

$$1 + G(s)K_p = 0 \Rightarrow s^3 + 2.89s^2 + 1.445s + 3.33s = 0$$

Solving for s we get :

$$s = -0.9404 \pm 0.7399j$$

Q 9. $G(s) = K(s+a)/s(s+3)(s+6)$



$$\theta_4 - \theta_3 - \theta_2 - \theta_1 = (2n+1)\pi$$

$$\tan^{-1}\left(\frac{100}{a-1}\right) - \left[\tan^{-1}\left(\frac{100}{5}\right) + \tan^{-1}\left(\frac{100}{2}\right) + \tan^{-1}\left(\frac{100}{1}\right) + \pi \right] = (2n+1)\pi$$

$$\tan^{-1}\left(\frac{100}{a-1}\right) - 266.56^\circ = (2n+1)\pi$$

$$\Rightarrow \tan^{-1}\left(\frac{100}{a-1}\right) - 266.56 = -180$$

$$\tan^{-1}\left(\frac{100}{a-1}\right) = 86.56$$

$$\frac{100}{a-1} = 16.635$$

$$a-1 = 6.011$$

$$\Rightarrow \boxed{a \approx 7}$$

Now $K = \frac{-1}{G(s)}$

$$= \frac{-s(s+3)(s+6)}{(s+7)} \quad \Big|_{s=-1+100j}$$

$$K = 9997$$

For $a=7$ and $K=9997$ the system will yield closed loop pair of poles at $-1 \pm 100j$ second order

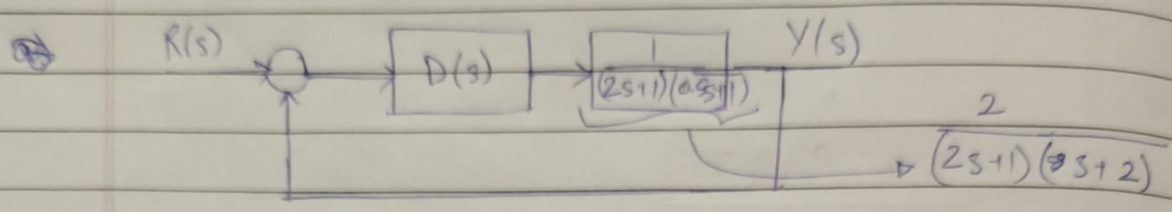
* For $K=9997$ and $a=7$

$$s(s+3)(s+6) + K(s+a) \approx (s+7)(s+1-100j)(s+1+100j)$$

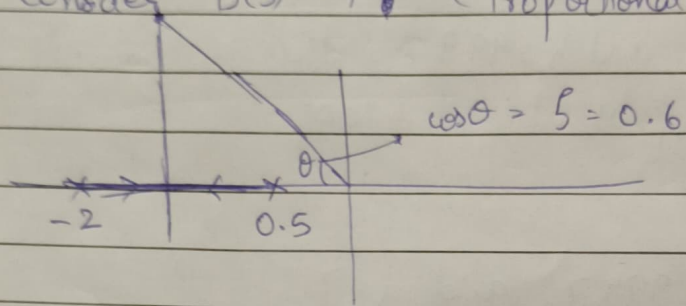
as $7 > 4 \times 1$ we can approximate the system as 2nd order.

Q10 Given $\zeta = 0.6$, $\zeta \omega_n = 0.75$

$K_p \rightarrow \infty$ (for 0 (ess) step)



a) Consider $D(s) = K_p$ (Proportional control)



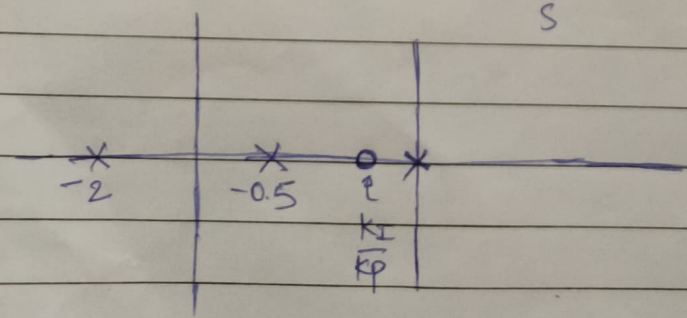
meeting point of asymptote = $\frac{(-2 + -0.5)}{2} = -1.25 = \sigma_{wm}$

Now as for $\zeta = 0.6$ the only possible time constant is $1/1.25$ which will not meet our requirements.

* even if the asymptotes would meet at 0.75 , the system will not be desirable as there will be some finite steady state error.

$\zeta = 2$
 $\frac{(2s+1)(s+2)}{s} = 1$

b) Now consider $D(s) = \frac{K_I}{s} + K_p \equiv K_p \left(\frac{s + \frac{K_I}{K_p}}{s} \right)$



Now we can make the ~~asymptotes meet at~~ by choosing appropriate value of K_I/K_P , as they would meet at

$$\sigma = -2.5 + \frac{K_P/K_I}{2}$$

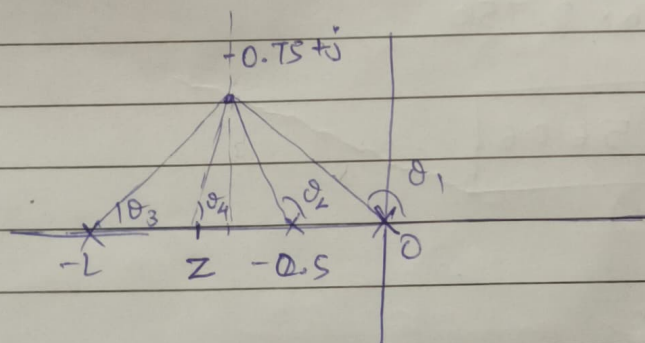
We can choose appropriate values of K_I/K_P , s.t. the required pole would lie on the root locus.

* Also as the zero is very close to origin the root locus will approximately the same, but now the steady state error will become zero as for ~~step~~ step input.

$$e_{ss} = \lim_{s \rightarrow 0} \frac{G(s)}{1+G(s)} = \lim_{s \rightarrow 0} \frac{1}{1 + \underbrace{(K_P + \frac{K_I}{s})}_{\rightarrow \infty} \underbrace{G(s)}_{\rightarrow 1}} = 0.$$

(c) ~~$-1.25 + \frac{K_P/K_I}{2} = -0.75$~~
 ~~$\frac{K_P}{K_I} = 1$~~ $\frac{K_P}{K_I} = 1$

required poles $-0.75 \pm j$



$$\Sigma \angle = \pi - \tan^{-1} \frac{1}{1.25}$$

$$\theta_4 - \theta_3 - \theta_2 - \theta_1 = (2n+1)\pi$$

$$\tan^{-1} \left(\frac{1}{2-0.75} \right) - \left[\tan^{-1} \left(\frac{1}{1.25} \right) + \pi - \tan^{-1} \left(\frac{1}{0.25} \right) + \pi - \tan^{-1} \left(\frac{1}{0.75} \right) \right] = (2n+1)\pi$$

$$\tan^{-1} \left(\frac{0.75}{z - 0.75} \right) = 269.566 = -180$$

$$\tan^{-1} \left(\frac{1}{z - 0.75} \right) = 89.566$$

$$\frac{1}{z - 0.75} = 132.0154$$

$$\boxed{z = 0.7576} = K_I / K_P$$

$$\text{Now } K_P = \frac{-s(s+1)(0.5s+1)}{(s+0.7576)}$$

$$s = 0.75 + j$$

$$\boxed{K_P = 2.06}$$

$$K_I = 2.06 \times 0.7576$$

$$\boxed{K_I = 1.5606}$$